

Morphological changes induced by buffer exchange during preparation of mRNA-lipid nanoparticles occur in a stepwise manner as pH is elevated

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Initial entrapment of nucleic acids in lipid nanoparticles (LNPs) is dependent on the use of ionizable cationic lipids, which draw nucleic acids into lipid particles at low pH in the presence of ethanol. Manufacturing of fully formed LNPs is completed by buffer exchange and removal of ethanol. We studied particle morphology at intermediate pH values during buffer exchange using an fluorescence resonance energy transfer (FRET) assay to indicate particle interactions, particle sizing, cryo-electron microscopy (cryo-EM), and small-angle X-ray scattering (SAXS). We compared LNPs formed by different ionizable lipids, including DLin-MC3-DMA, ALC-0315, and SM-102, used, respectively, in the Food and Drug Administration (FDA)-approved products, Onpattro, Comirnaty, and Spikevax. FRET and cryo-EM studies confirmed that particle interaction and fusion occurred during buffer exchange. By dialyzing LNPs in various buffers, we found that stable particles with bleb-like protrusions were formed at intermediate pH (i.e., pH 5.5). Fusion and particle growth occurred at higher pH values for SM-102 LNPs, reflecting the higher pKa of SM-102. SAXS profiles showed that, when fully formed at the final pH of 7.4, MC3 and ALC-0315 LNPs had lost bilayer-like structures, which were present after particle formation at pH 4. In contrast, LNPs produced with 1,2-dioleoyloxy-3-dimethylaminopropane (DODMA) and SM-102 retained some bilayer-like structures at pH 7.4.

INTRODUCTION

Lipid nanoparticles (LNPs) are currently the leading technology for non-viral delivery of nucleic acids. The first Food and Drug Administration (FDA)-approved pharmaceutical product making use of LNP technology was Onpattro,¹ which was developed for intravenous delivery of the small interfering RNA (siRNA) drug patisiran. Since 2020, LNPs have been used to deliver mRNA in the COVID-19 vaccines, Comirnaty² and Spikevax,³ which encode variants of the SARS-CoV-2 spike protein. The entrapment of mRNA by LNPs is now a routine procedure, in practice in many academic and industrial research laboratories, although it is a comparatively young technology. Nevertheless, there is much to learn about the

morphological changes that occur during the standard manufacturing process, the morphology of the finished product, and the fate of LNPs and their nucleic acid payloads after parenteral administration.

The standard method for manufacture of mRNA-LNP is to mix a solution of mRNA in acidic buffer (usually at pH 4) with an ethanol solution of lipids, typically at a 3: 1 (mRNA: lipids) flow ratio.⁴ Manipulation of mixing methods and manufacture conditions often impact on the formation and morphology of the LNP products. Microfluidic mixing methods have been reported to produce better control over particle size and improved mRNA encapsulation efficiency when compared to turbulent mixing,⁵ though it is important to recognize that microfluidic mixing is not a scalable process in the context of large-scale manufacturing. Particles formed initially at pH 4 are subsequently subjected to buffer exchange to remove ethanol and adjusted to physiological pH. Adjustment of buffering conditions and salt concentration may induce bleb-like protrusions in normal manufacturing procedures.⁶ The standard Onpattro formulation includes an ionizable lipid, cholesterol, distearoylphosphatidyl choline, and a PEGylated lipid in the ratio of 50:38.5:10:1.5 defined as molar percentage. This combination produces LNPs with dense amorphous cores. Alternative combinations of lipids or composition may form aggregated inverted micelles or multilamellar structures in the LNP cores.^{7,8} Typically, the mRNA is mixed with lipids such that the cationic-nitrogen-to-anionic-phosphate ratio (N/P ratio) is approximately 5–6. Preparation of LNPs using lower N/P ratios formed larger particles with reduced bilayer structure and was reported to affect transfection efficiency.⁹

During buffer exchange, there is substantial rearrangement of the primary particles to form the finished LNP product. Morphological changes occur because the majority of the ionizable lipid molecules

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are deprotonated during buffer exchange. Previously, Semple and colleagues identified that the ionizable lipids and phospholipids in a typical LNP formed a curved, core structure at the manufacturing pH (pH 4).¹⁰ Using cryo-electron microscopy (cryo-EM), Kulkarni and colleagues studied the events that occur on buffer exchange and suggested that internal structural and morphological changes occur that allow the ionizable lipids to be re-located during neutralization.⁹ This may result in “particle fusion” and increase in mean size distribution of LNPs as the pH is changed from the initial manufacturing pH (pH 4) to the pH of the finished product (typically pH 7.4). Later, a fluorescence resonance energy transfer (FRET) method was established by the same group,¹¹ which was thought to confirm that particle fusion was widespread. Recent research using small-angle neutron scattering analysis also revealed that the acidic buffer used during formation of primary particles had effects on size distribution and water content of LNPs.¹²

Previous studies have focused on the differences in morphology between particles formed at pH 4 and those present in the final LNP product at pH 7.4. In this study, we used cryo-EM, small-angle X-ray scattering (SAXS), and the FRET fusion assay to investigate whether morphological changes that occur at intermediate pH ranges during buffer exchange are transient or durable, i.e., whether or not stable intermediate structures are formed under different conditions of pH, and whether intermediate structures formed are influenced by the pKa of each ionizable lipid. We posit that improved understanding of the kinetics of morphological change will assist in understanding how dialysis conditions or diafiltration protocols could influence the finished LNP product. We compared LNPs made with the ionizable lipid used in Onpattro, DLin-MC3-DMA (MC3), with LNPs made using the lipids used in the mRNA SARS-CoV-2 vaccines, and with an early-generation lipid, 1,2-dioleoyloxy-3-dimethylaminopropane (DODMA).¹³ We found that morphological changes could be detected using all three experimental methods and that changes occurred over a pH range that was dependent on the practical pKa of each ionizable lipid. SAXS studies indicated that bilayer-like structures were present at low pH but were gradually lost if LNPs were dialyzed against buffers at higher pH. Cryo-EM studies suggested that particle fusion and condensation during buffer exchange are stepwise processes that are pH dependent, which take place over a critical intermediate pH range of approximately one pH unit. Morphological changes in LNPs prepared using SM-102, the ionizable lipid used in Spikevax, occurred at higher pH, reflecting the higher pKa of SM-102. SAXS studies suggested that some bilayer-like structures remained in SM-102 LNPs after buffer exchange to pH 7.4, which can be explained by the presence of a significant proportion of protonated SM-102 at pH 7.4. Our studies established a suite of techniques that can be used to estimate the effective practical pKa within the LNP product. These techniques offer formulators the opportunity to select buffers that are appropriate for manufacture of LNPs in accordance with the apparent pKa of the ionizable lipid selected.

RESULTS

Morphological changes occurring during production of MC3 LNPs

To investigate particle fusion during buffer exchange, we used the FRET assay first reported by Kulkarni et al.¹¹ An FRET donor 3,3'-dioctadecyloxycarbocyanine perchlorate (DiO) (excitation maximum [ex.] 484 nm and emission maximum [em.] 501 nm) and an FRET acceptor 1,1'-dioctadecyl-3,3',3'-tetramethylindocarbocyanine perchlorate (DiI) (ex. 549 nm and em. 565 nm) were formulated in separate populations of empty LNPs; then, the two products were combined in a 1:1 ratio and diluted 10-fold in an appropriate buffer to change the pH of the continuous aqueous phase. UV absorption spectra of the LNPs confirmed that the Di-fluorophores were incorporated into the LNPs, were not lost during dialysis, and were incorporated at equivalent concentration in the two set of LNPs (Figure S1). The two lipophilic fluorophores acted as an FRET pair when in close proximity. In agreement with published data, after mixing LNPs at pH 7.4, but not at pH 4.0, an FRET signal was evident at 565 nm, after excitation at 470 nm. The FRET signal at 565 nm developed within a few minutes after mixing, reaching a maximum with 15 min (Figure S2). After 15 min, fluorescence spectra at pH 4 and pH 7.4 of individual and mixed LNP populations after excitation at 470 nm (Figure 1A) were essentially identical to spectra reported by Kulkarni et al.¹¹ indicating that aggregation and/or fusion occurred after deprotonation of MC3 at pH 7.4, but not when MC3 was protonated at pH 4.0. To explore whether the FRET signal was produced at intermediate pH values, we diluted mixtures of the two LNP populations in buffers to produce final pH values at 0.25 pH unit intervals between pH 5.00 and 6.50 and collected the spectra shown in Figure 1B after excitation at 470 nm. Figure 1C shows the relative fluorescence of these mixtures at 505 nm (emission from DiO) and 565 nm (the FRET signal) at specific pH values. The relative fluorescence at 565 nm at pH 4 is equivalent to the sum of the emissions due to the individual DiO and DiI dyes. At pH 5, additional fluorescence due to the FRET was evident, and, as the pH was raised to 5.25 and 5.50, there was a stepwise increase in FRET, which appeared to reach a maximum at pH 5.50. The FRET signal was unchanged between pH 5.50 and 7.40. The fluorescence intensity at 505 nm was reduced as the pH was raised suggesting that, by pH 5.75, approximately 70% of the DiO was engaged in FRET activity. The mean particle size and polydispersity index (PDI) of LNP mixtures were estimated at each pH value by dynamic light scattering (DLS) (Figure 1D). Particles produced at pH 4 (DiO, DiI, and the mixtures at pH 4) were small, heterogeneous mixtures of particles, with Z-average diameters of 30–40 nm, which included particles in the 15 nm range as shown by the predominant volume average diameter (i.e., the mean particle diameter of the main peak indicated by DLS). The PDI values at pH 4.0 were >0.5 confirming that, in accordance with published data, primary particles produced at pH 4 were heterogeneous and included some smaller particles that resembled unilamellar liposomes (see also Figure 2B). When the pH was increased to 5.00, the Z-average diameter increased and at pH 5.25 and 5.50 reached a steady state with volume main peak of 40–50 nm

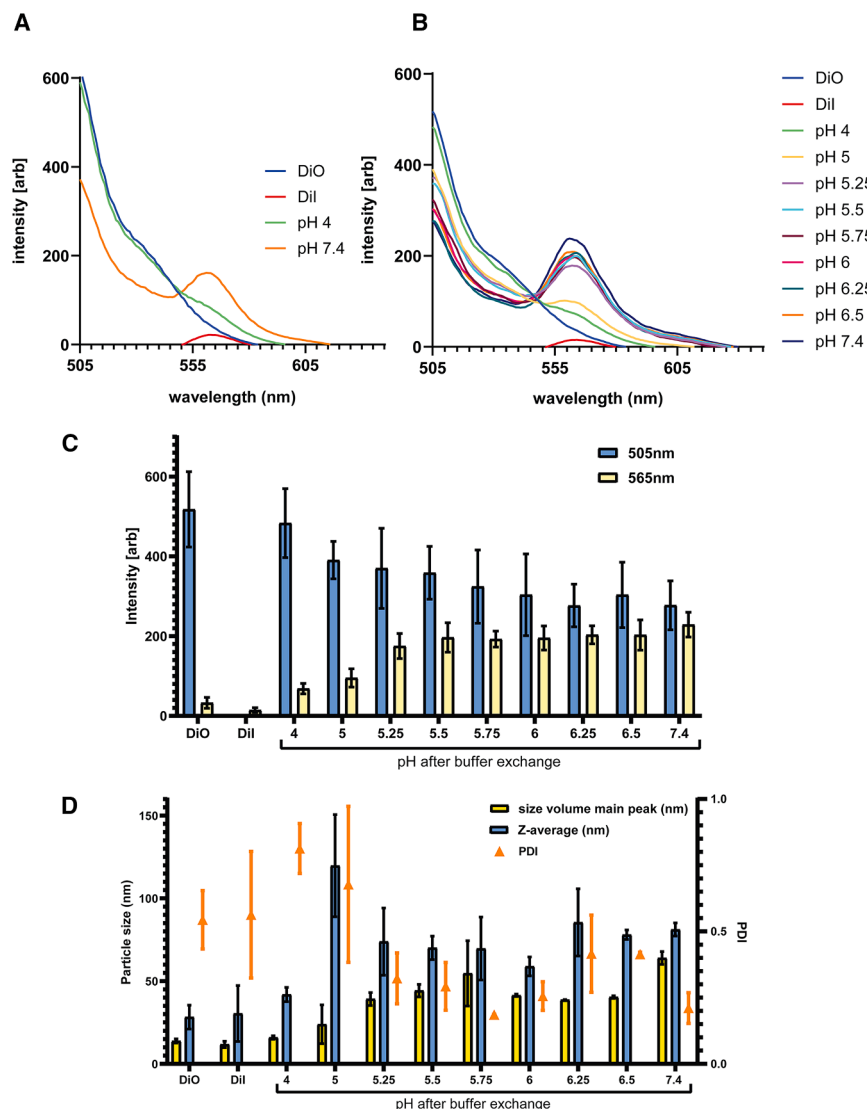


Figure 1. FRET investigation of particle fusion during preparation of MC3 LNP formulations

All formulations were empty LNPs consisting of MC3/DSPC/cholesterol/DMG-PEG 2000 (mole ratio = 50/10/38.5/1.5) spiked with either DiO or Dil. (A) Relative fluorescence spectra of empty LNP-DiO particles excited at 470 nm at pH 4 (labeled DiO, shown in blue), LNP-Dil particles excited at 470 nm at pH 4 (labeled Dil, shown in red), and mixed LNP-DiO and LNP-Dil particles excited at 470 nm after mixing at pH 4 (shown in green) or after buffer exchange to pH 7.4 (shown in orange). FRET can be observed by additional emitted light at 565 nm. (B) Representative fluorescence spectra after dilution of mixed LNP-DiO and LNP-Dil particles with various buffers to generate pH conditions ranging from pH 4 to pH 7.4. (C) Fluorescence intensities of replicate samples prepared as in (B) at 505 nm (the emission maximum of DiO dye) or 565 nm (the emission maximum of Dil dye) after excitation at 470 nm (mean and SD, $n = 5$). (D) Particle size distribution and polydispersity index (PDI) of MC3 LNP-DiO/LNP-Dil mixtures at each pH value. Particle size was estimated by dynamic light scattering (DLS) and is shown as volume average particle diameter and Z-average particle diameter (mean and SD, $n = 5$).

and Z-average of 70–80 nm over the pH range from 5.25 to 6.50. There was a decrease in PDI at pH 5.25 and above, suggesting that, during particle fusion, the LNP population became more uniform. Particle size and polydispersity of each LNP mixture were also investigated using nanoparticle tracking analysis (NTA). This technique confirmed the particle size distribution (data not shown) and identified the presence of a limited number of large particles in mixtures with high PDI.

Figure 2 shows SAXS and cryo-EM data for LNPs loaded with mRNA, initially formed using standard conditions, mixing acetate buffer at pH 4, in the presence of 25% ethanol. The LNPs were subsequently dialyzed against buffers at the pH values shown in Figure 2 and analyzed within 14 days. Representative cryo-EM images at various pH values are shown in Figure S3. SAXS profiles (Figure 2) indicated that at pH 4.00 there was a distinct but broad diffraction peak at a q value

of approximately $0.11 \text{ \AA}^{-1}$, suggesting structural organization with the LNPs. This peak was observed at a similar q value at all pH values between 4.00 and 6.50 but was much reduced in intensity or was absent at pH 7.40. The peaks observed at $0.11 \text{ \AA}^{-1}$ corresponded to a d -spacing between planes of 5–6 nm ($d = 2\pi/q$), which is consistent with the spacing between head groups in lipid bilayers, although the SAXS peak was broader than what would be expected for lamellar bilayers, such as those reported for liposomes produced with pure phospholipids.¹⁴

Multiple orders of diffraction peaks (spacing ratio 1:2:3:4:5 ...) usually observed with multilamellar lipid vesicles were not observed. Due to the inherent polydispersity in the particle size distributions measured through DLS (Figure 1), analysis of the overall particle structures utilizing the low q scattering data was not undertaken as part of this study. Cryo-EM images of LNPs dialyzed in different pH buffers (Figures 2B and S3) showed that at pH 4 the primary particles included distorted electron-dense LNPs as well as unilamellar liposomes. At pH 7.4, fully formed LNPs were consistently spherical, dense particles. At intermediate pH values such as 5.5, the morphology of particles was consistent with partially formed LNPs, often containing dense regions with associated vesicular blebs as shown previously,¹⁵ suggesting that aggregation and fusion of MC3 particles to form a finished LNP product were incomplete at pH 5.5. Interestingly, the partially formed LNPs were stable at pH 5.5 suggesting that the speed at which buffer exchange takes place may not have a critical influence on the morphology of the finished product.

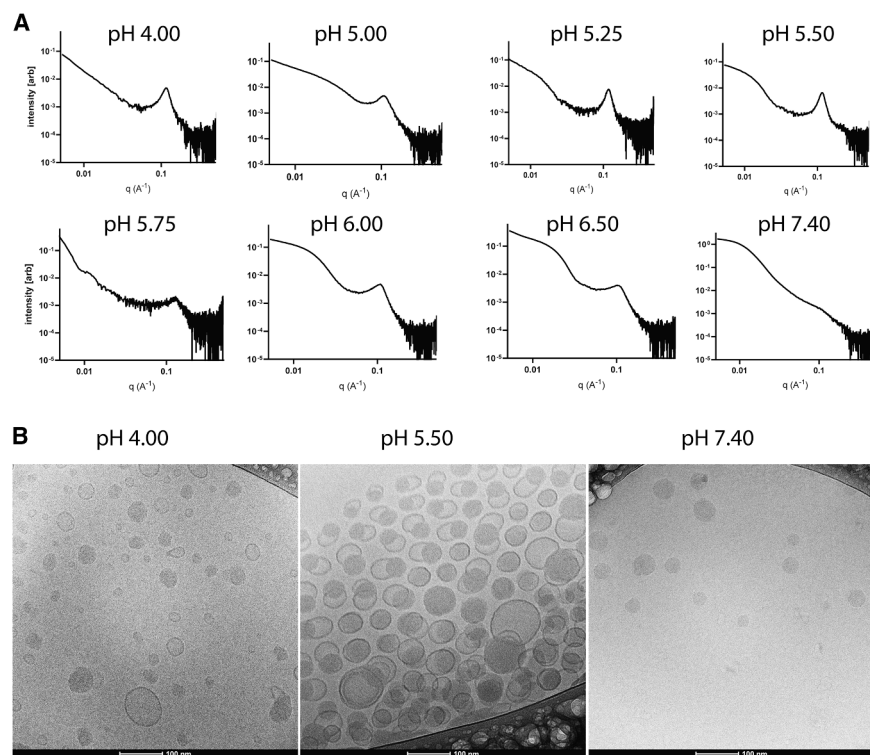


Figure 2. Structural changes in MC3 mRNA-LNP formulations induced by buffer exchange

(A) Small-angle X-ray scattering (SAXS) profiles of LNPs consisting of MC3/DSPC/cholesterol/DMG-PEG 2000 (mole ratio = 50/10/38.5/1.5) and mRNA at N/P ratio = 6, after dialysis, in each case to remove ethanol and result in an aqueous continuous phase with the pH values shown in each panel. (B) Representative cryo-EM images of mRNA-LNPs after dialysis as in (A) to produce a final pH of 4, 5.5, or 7.4.

et al.¹⁹ reported a pKa of 6.1, in common with the value 6.09, reported in Acuitas Therapeutics' patent WO2017075531A1. We found that DODMA had a pKa value (6.41), which was close to ALC-0315, but SM-102 had a higher pKa (6.70), consistent with a published value for SM-102.²⁰ Raw data determined during pKa determination for three batches of each LNP used in this study are shown in Figure S5. All the ionizable lipids had measured pKa values within 0.3 pKa units of 6.5, which has been associated with good biological activity, though our TNS assay data suggest that the difference between the pKa values of MC3 and

SM-102 is close to 0.5 units. Thus, morphological changes might be expected to take place at subtly different pH values but within 0.5 pH units of each other for the lipids used in this study.

Morphological changes occurring during production of LNPs with alternative ionizable lipids

A specific objective of this study was to compare morphological changes during dialysis of LNPs produced with MC3 with the equivalent effects on LNPs produced using alternative ionizable lipids and to investigate whether any differences with the various LNPs were related to the pKa of each ionizable lipid. We used a consistent formulation for LNPs (based on mole % of each lipid component) and changed just the identity of the ionizable lipid. Chemical structures of the lipids used in this study are shown in Figure S4. We compared the ionizable lipid MC3 (used in Onpatro) with the lipids used in Spikevax (SM-102), Comirnaty (ALC-0315), and a structurally related pair of older lipids to investigate the difference between the ionizable lipid DODMA and its quaternary equivalent DOTMA. We determined the apparent pKa values of each lipid using the TNS titration method (Table 1). The TNS method proved to be reproducible in our hands, but the pKa values were not identical to published values. This could be explained by the experimental conditions used, such as LNP and TNS concentrations, which were not consistent across published studies, a factor that has been discussed previously by Carrasco and colleagues.¹⁶ Published pKa values for specific lipids determined by the TNS method are usually within 0.3 pH units but vary within this range. Our pKa values were determined using consistent experimental conditions, which we suggest make comparisons between lipids more robust. We found that MC3 had the lowest pKa (6.26) of the lipids we investigated. Values ranging from 6.31 to 6.57 have been reported previously.¹⁶ ALC-0315 had a higher pKa (6.36), in agreement with a recent published value of 6.38,¹⁸ though McMillan

Figure 3 shows data from FRET assays and particle sizing assays of mixtures of DiO- and DiI-labeled LNPs made using either ALC-0315 or SM-102, in analogy with the data for MC3 shown in Figure 1. Spectra collected for mixtures held at each pH after excitation at 470 nm are shown in Figures 3A and 3B. There was strong resemblance with spectra collected for the MC3 particles (Figure 1B), but closer inspection of the fluorescence intensities at 505 and 565 nm (Figures 3C and 3D) reveal that the FRET signal changes over subtly different pH ranges. The FRET signal for ALC-0315 particles (Figure 3C) increases primarily over a narrow pH range reaching close to the maximum FRET signal between 4.75 and 5.00. In contrast for SM-102 particles (Figure 3D), the significant changes occur over a higher pH range, reaching a maximum FRET signal at pH 5.25. The proportional FRET signal intensity (565 nm/505 nm) from mixtures of particles was the highest for ALC-0315 when all the LNP formulations were compared (see for a clear comparison Figures 3C for ALC-0315 versus Figure 5C for DODMA). This implies that a higher proportion of DiO molecules are in close proximity to DiI molecules, indicating that the fusion phenomenon might involve an unusually high proportion of particles during buffer exchange of ALC-0315 LNPs. The proportional FRET signal intensity (565 nm/505 nm) could potentially be used to estimate the practical pKa of each ionizable lipid within an LNP. The growth in particle diameter and reduction in PDI for SM-102 (Figure 3F), seen most clearly in the volume

Table 1. pKa values of ionizable lipids

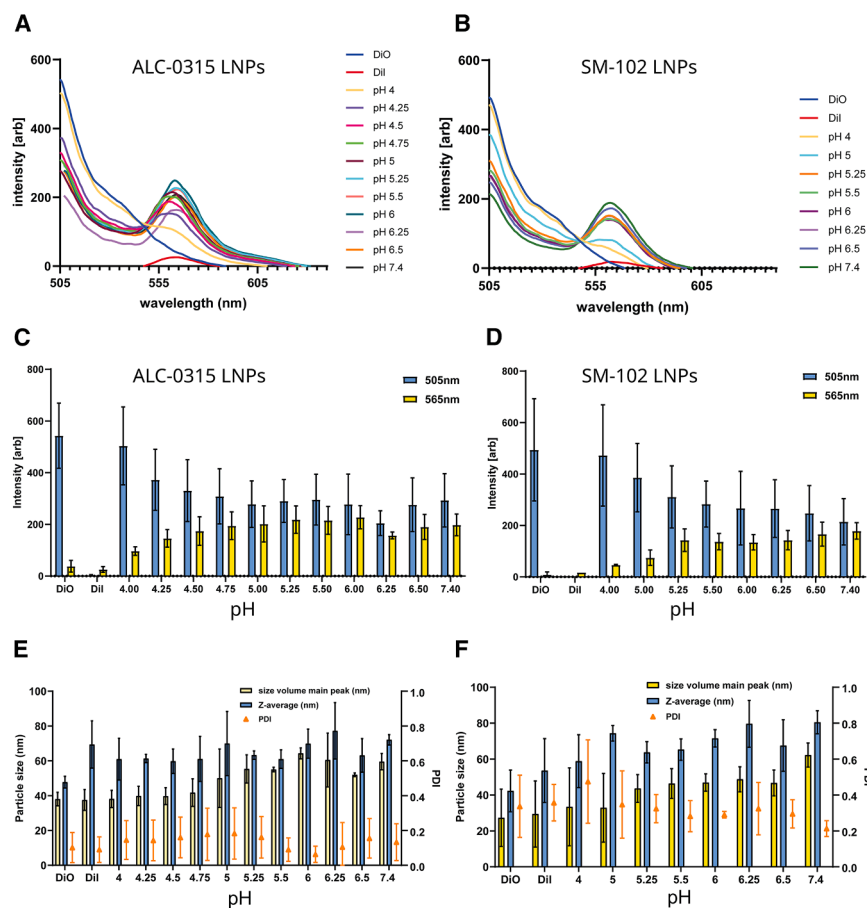
Lipid	DODMA	MC3	ALC-0315	SM-102
Mean pKa from this study ($\pm$ SD) ($n = 3$)	6.41 (0.07)	6.26 (0.06)	6.36 (0.06)	6.70 (0.04)
Published pKa values	6.59 Carrasco et al. ¹⁶	6.44 Jayaraman et al. ¹⁷	6.38 Kamanzi et al. ¹⁸ 6.1 McMillan et al. ¹⁹	6.68 Hassett et al. ²⁰

average mean diameter, stabilized at the pH of 5.25, corresponding to the pH when the FRET signal reached a maximum. Particle size data for ALC-0315 (Figure 3E) were more variable, though in general the trend toward larger particles with lower PDI at higher pH values was evident at pH 5.50 and higher pH values.

Figure 4 shows the influence of pH on SAXS profiles of mRNA-LNP particles produced with four different cationic lipids. SAXS profiles at several intermediate pH values are shown in Figures S6–S9. The ionizable lipids ALC-0315 (Figures 4A and S6), SM-102 (Figures 4B and S7), and DODMA (Figures 4C and S8) all exhibited a broad diffraction peak at q values close to $0.11 \text{ \AA}^{-1}$, but the q values were

subtly different resulting from different d-spacings (discussed in more detail below). The diffraction peaks were still evident at pH 6.5, but at pH 7.4 the peak for ALC-0315 (in common with MC3) was greatly reduced in intensity. For SM-102 and DODMA (Figures 4B and 4C) LNPs, the diffraction peak around $q = 0.11 \text{ \AA}^{-1}$ was reduced but still evident at pH 7.4, reflecting the higher pKa values of these two lipids. This implies that a degree of order is still present for SM-102 and DODMA LNPs at pH 7.40, which can be explained by incomplete deprotonation of these lipids at pH 7.40. In contrast to ionizable DODMA, its quaternary counterpart, DOTMA, exhibited more than one broad peak in the SAXS profile, at pH values above 5.50. Both peaks remained at all pH values between 5.5 and 7.4 (Figures 4D and S9), consistent with the expectation that the ionized state of DOTMA was independent of pH. The presence of two broad peaks in this region of the scattering profiles is consistent with the formation of LNP-associated bilayer-like structures and an independent population of DOTMA micelles or vesicles.

The FRET signal after mixing of DiO- and DiI-labeled DODMA or mixing of DiO- and DiI-labeled DOTMA particles is shown in Figure 5. For DODMA, the FRET signal reached a low peak at pH 5.25–5.50, though a further increase in FRET signal was evident

**Figure 3. FRET investigation of particle fusion during preparation of ALC-0315 or SM-102 LNP formulations**

All formulations were empty LNPs consisting of ionizable lipid/DSPC/cholesterol/DMG-PEG 2000 (mole ratio = 50/10/38.5/1.5) spiked with either DiO or DiI. The experiments are analogous with MC3 experiments shown in Figure 1. Representative fluorescence spectra after dilution of mixed LNP-DiO and LNP-DiI particles with various buffers to generate pH conditions ranging from pH 4 to pH 7.4, for ALC-0315 LNPs (A) or SM-102 LNPs (B). Fluorescence intensities of replicate samples at 505 nm (the emission maximum of DiO dye) or 565 nm (the emission maximum of DiI dye) after excitation at 470 nm (mean and SD, $n = 5$), for ALC-0315 LNPs (C) or SM-102 LNPs (D). Particle size distribution and polydispersity index (PDI) of LNP-DiO/LNP-DiI mixtures at each pH value, for ALC-0315 LNPs (E) or SM-102 LNPs (F). Particle size was estimated by DLS and is shown as volume average particle diameter and Z-average particle diameter (mean and SD, $n = 5$).

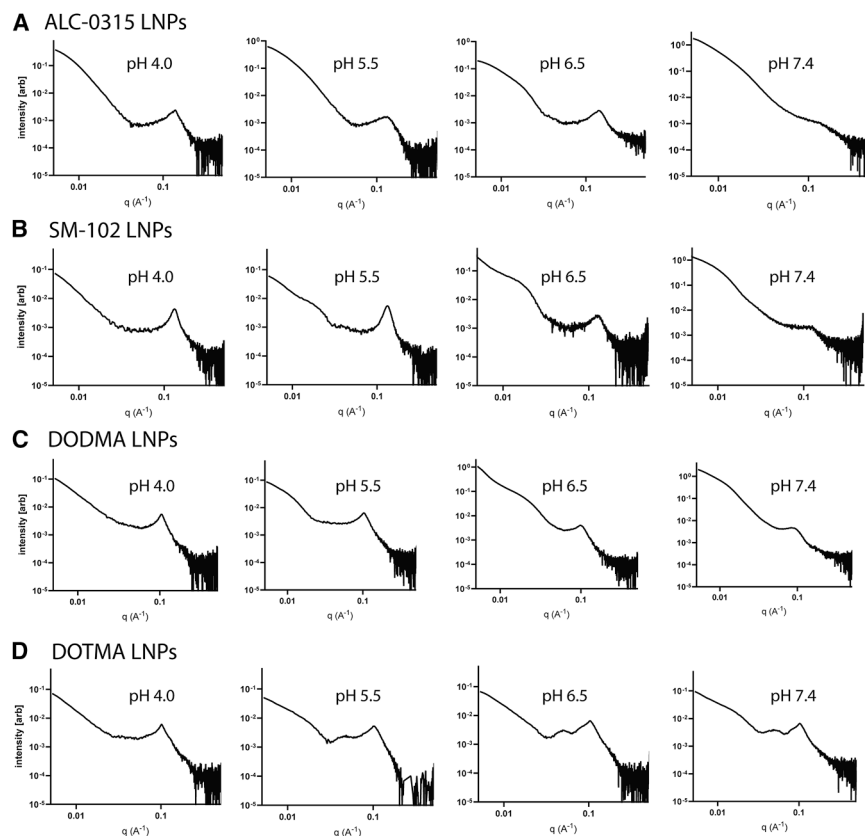


Figure 4. Small-angle X-ray scattering profiles of LNPs at selected pH values

mRNA-LNPs, consisting of cationic lipid/DSPC/cholesterol/DMG-PEG 2000 (mole ratio = 50/10/38.5/1.5) and mRNA at N/P ratio = 6, were initially formed in a standard manner at pH 4 in the presence of 25% ethanol and then dialyzed to remove ethanol and produce an aqueous phase with final pH of 4, 5.5, 6.5, or 7.4. Representative SAXS profiles obtained at each pH are shown for ALC-0315 (A), SM-102 (B), DODMA (C), and DOTMA (D).

between 6.50 and 7.40. Z-average particle size increased steadily between pH 5.00 and 6.25, consistent with the expectation of particle fusion as DODMA was deprotonated. PDI was unchanged between pH 5.00 and 7.40. FRET data and SAXS data combined for DODMA suggested that particle fusion to form amorphous core may have been incomplete at pH 7.40 for DODMA LNPs, similar to SM-102 LNPs: both of these ionizable lipids would have been incompletely deprotonated at pH 7.40. When Di-dye-labeled DOTMA LNPs were mixed, a FRET signal was evident which reached a maximum at pH values between 4.00 and 5.50, suggesting that, despite the permanent cationic charge of DOTMA, the two dyes were present in close proximity after the LNPs were mixed. LNP Z-average particle diameter and PDI were somewhat variable, but, in common with LNPs prepared with ionizable lipids, there appeared to be modest particle growth after mixing and buffer exchange.^{21,22} The extent of particle growth does not indicate extensive aggregation, but the FRET signal suggests that a degree of contact between DiO- and DiI-labeled LNPs occurred on mixing, even at pH 4.00.

Analysis of the structural order in LNPs from SAXS data

The magnitude of the scattering vector q allows estimation of the distance between repeating planes of structure within the particles from the position of the broad diffraction peaks observed ($d = 2\pi/q$) in the scattering profiles of the LNP dispersions. Values of d-spacing were generally within the range 4.5–6.5 nm (Figure 6), which was consis-

tent with the spacing between lipid head groups in bilayer-like structures. For LNPs that included 50 mole% ionizable lipids, when the d-spacing was plotted against pH, we observed a reduction in d-spacing at approximately pH 5, followed by an overall increase in d-spacing as the pH was increased further. For LNPs produced with DODMA or SM-102, which still exhibited the broad diffraction peak at pH 7.40, the d-spacing was observed to increase between 6.50 and 7.40. d-spacing for LNPs produced using either of the two branched ionizable lipids, ALC-0315 and SM-102 (Figure S4), was 1 to 1.5 nm shorter than those estimated for other LNPs, suggesting that, when ALC-0315 and SM-102 were protonated and incorporated into bilayer structures, the bilayers produced were thinner. MC3 LNPs were found to have bilayers approximately 1 nm thicker than bilayers produced in the presence of branched lipids. d-spacing was the thickest at 6.0–6.2 nm for the structural analogs, DOTMA and DODMA, and was consistent over the pH range 4.00–6.00. But for the quaternary cationic lipid, DOTMA, there was no change in d-spacing across a pH range up to 7.40, consistent with the expectation that it was permanently positively charged, irrespective of pH. In contrast, for DODMA, in common with SM-102, the d-spacing increased between pH values of 6.00 and 7.40.

DISCUSSION

LNPs formulated with ionizable cationic lipids have been used to encapsulate a variety of biologically active nucleic acids,²³ i.e., plasmid DNA,²⁴ siRNA,⁴ or mRNA.²⁵ The chemistry of ionizable lipids used to produce LNPs is known to affect the biological activity of the nucleic acid product, both *in vitro* and *in vivo*.^{26,27} The internal structure of LNPs is also likely to be determined by the chemistry of ionizable lipids and the mole fraction of each lipid used, but analysis of the differences between LNP formulations has been limited by difficulties in establishing methods for detailed analysis, made difficult in particular by the amorphous nature of the core of LNPs. More detailed understanding of the internal structure and surface chemistry is needed and could potentially explain the relationships between LNP formulation, biodistribution, cellular uptake, intracellular trafficking, and biological activity.^{28–31}

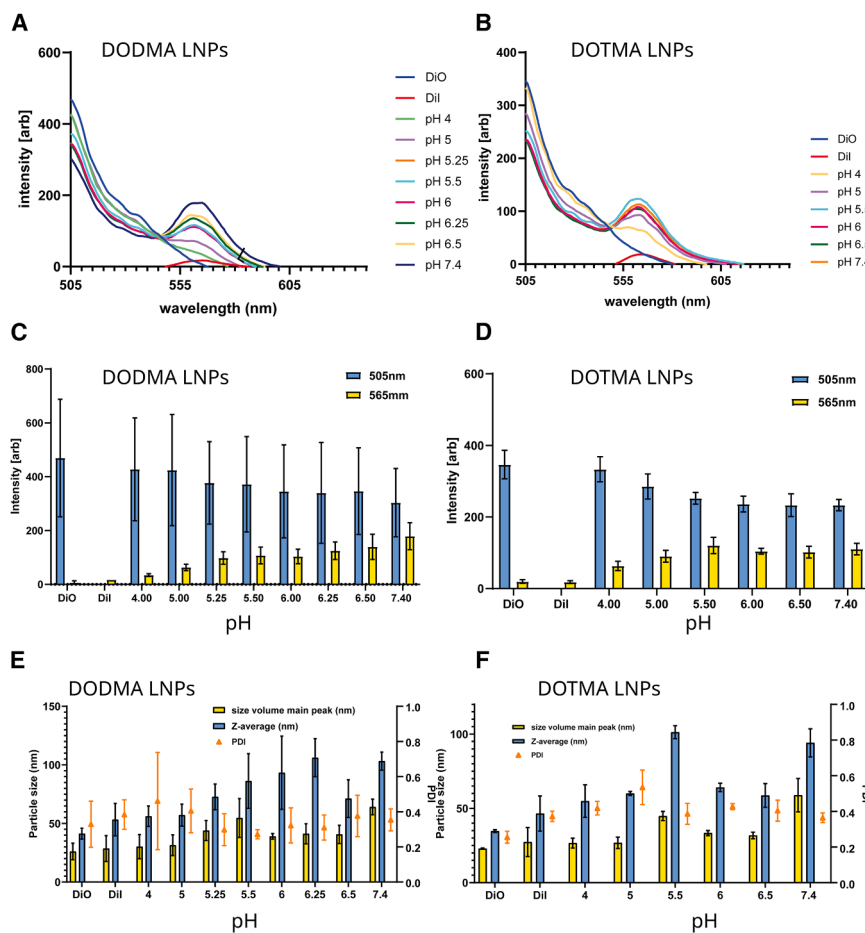


Figure 5. FRET investigation of particle fusion during preparation of DODMA or DOTMA LNP formulations

All formulations were empty LNPs consisting of cationic lipid/DSPC/cholesterol/DMG-PEG 2000 (mole ratio = 50/10/38.5/1.5) spiked with either DiO or DiI. The experiments are analogous with MC3 experiments shown in Figure 1. Representative fluorescence spectra after dilution of mixed LNP-DiO and LNP-DiI particles with various buffers to generate pH conditions ranging from pH 4 to pH 7.4, for DODMA LNPs (A) or DOTMA LNPs (B). Fluorescence intensities of replicate samples at 505 nm (the emission maximum of DiO dye) or 565 nm (the emission maximum of DiI dye) after excitation at 470 nm (mean and SD, $n = 5$), for DODMA LNPs (C) or DOTMA LNPs (D). Particle size distribution and polydispersity index (PDI) of LNP-DiO/LNP-DiI mixtures at each pH value, for DODMA LNPs (E) or DOTMA LNPs (F). Particle size was estimated by DLS and is shown as volume average particle diameter and Z-average particle diameter (mean and SD, $n = 5$).

experimental evidence that particle fusion and rearrangement occur in a stepwise manner as the ionizable lipid is deprotonated in the intermediate pH range during buffer exchange, resulting in consolidation of empty liposome-like particles, observed in the electron microscopy images, to form dense LNPs containing mRNA trapped in a substantially amorphous core. Our data extend the understanding of how structural changes occur during buffer exchange, i.e., by stepwise changes that can be stably isolated at intermediate pH.

The current study is focused on the role of ionizable lipids on formation of LNPs and the morphological changes that occur during their manufacture, in particular during the buffer exchange process. We adopted the FRET study based on the publication by Kulkarni et al.¹¹ and used this method to investigate whether particle fusion or rearrangement occurred in a stepwise manner as the pH was raised. We combined the FRET study with analysis of changes in particle size and PDI during buffer exchange to further confirm the critical pH range at which particle aggregation and/or fusion occurred. Particle size measurements during the FRET studies were made within 4 hours of buffer exchange by dilution but were generally unchanged over 24 h. We observed that the FRET signal did indeed increase in a stepwise manner and that particle diameter increased as the ionizable lipid was deprotonated, consistent with the increase in mean particle diameter caused by fusion. By capturing particles at intermediate pH values, we were able to investigate the morphological changes that occur sequentially as pH is elevated, using both SAXS and cryo-EM. Our observations from cryo-EM studies were in broad agreement with the conceptual diagram of morphological changes in LNPs, during buffer exchange, proposed in a previous study, which compared features observed at pH 4 versus those observed in the finished LNPs at pH 7.4.⁹ Here, we provide new

experimental evidence that particle fusion and rearrangement occur in a stepwise manner as the ionizable lipid is deprotonated in the intermediate pH range during buffer exchange, resulting in consolidation of empty liposome-like particles, observed in the electron microscopy images, to form dense LNPs containing mRNA trapped in a substantially amorphous core. Our data extend the understanding of how structural changes occur during buffer exchange, i.e., by stepwise changes that can be stably isolated at intermediate pH. We also confirm that mature LNPs (at pH 7.4) are formed by fusion of electron-dense cores with less electron-dense vesicular structures.³² Cryo-EM images after dialysis at intermediate pH 5.5 showed that many particles formed in the presence of mRNA (at N/P ratio = 6) were in a partially fused state, with vesicular bilayer-like blebs fused to electron-dense regions. These intermediate states were relatively stable and were only resolved to yield fully formed LNPs by dialysis in buffer at pH 7.4. The process of morphological change is shown schematically in Figure 7, in general agreement with previous studies that compared LNPs initially formed at pH 4 with finished LNPs at pH 7.4.^{9,11,12,33} It should be noted that the LNPs used for SAXS and cryo-EM studies were produced using a relatively slow static dialysis technique for buffer exchange. Under manufacturing conditions, faster tangential flow filtration is generally used, which could result in differences in the morphology of finished LNPs.

Three of the ionizable lipids used in this study, MC3, ALC-0315, and SM-102, were chosen because they have been used in FDA-approved products and are now widely used in contemporary LNP studies. The fourth ionizable lipid, DODMA, is an example of an earlier ionizable

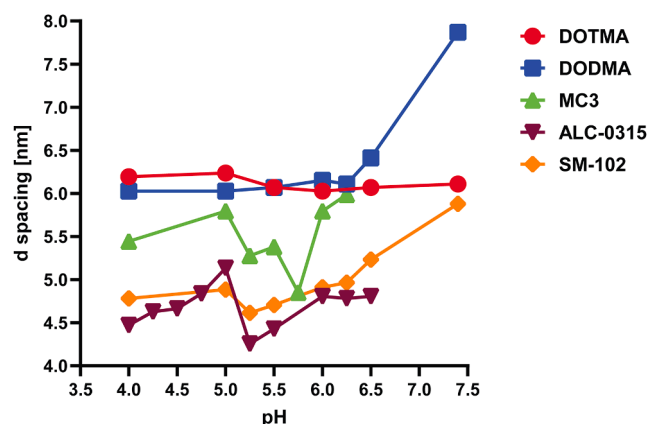


Figure 6. Estimates of d-spacing obtained from SAXS profiles

Estimates of the d-spacing from the broad diffraction peaks in the SAXS profiles for each LNP as a function of pH.

lipid that is the dimethylamino analog of the well-known quaternary lipid DOTMA.¹³ The pKa values of the four ionizable lipids are relatively close, which made it difficult to identify clear differences in morphology at each pH studied. The pKa of the ionizable lipid SM-102, which had the highest measured pKa of 6.70, theoretically would be expected to be proportionally more protonated at each pH than MC3 or ALC-0315. This resulted in the development of the FRET signal at higher pH. SM-102 and DODMA LNPs, with higher pKa values, contained a higher proportion of protonated lipids at pH 7.40 (e.g., SM-102 should theoretically be 20% protonated at pH 7.4). This was evident in our SAXS data, which indicated that some bilayer structures were still present in SM-102 and DODMA LNPs at pH 7.40, in contrast to the SAXS profiles of MC3 and ALC-0315 LNPs, which suggested that the latter LNPs had lost the majority of their bilayer structures. SAXS analysis was unequivocal in demonstrating that DOTMA LNPs were unaffected by pH in relation to bilayer content and d-spacing above pH 5, but curiously DOTMA LNPs produced an FRET signal after buffer exchange. To investigate this unexpected result, we used cryo-EM to establish whether fusion can occur with DOTMA LNPs. Figure S10 shows three representative images of DOTMA formulations at each of three pH values: pH 4.0, 5.5, and 7.4. The cryo-EM images confirmed that DOTMA formulations contained a mixture of electron-dense and “empty” liposome-like particles at all pH values, suggesting that fusion to form finished LNPs, typical of formulations prepared with ionizable lipids, did not occur. To investigate whether FRET signals occurred with other quaternary cationic lipids, we carried out FRET studies using formulations in which the ionizable lipid was replaced with DOTAP. Figure S11 shows clearly that the FRET signal was produced when the pH was changed to 5.5 or 7.4. Our observations with DOTMA and DOTAP suggest that the FRET signal is produced by particle contact, which can occur without subsequent irreversible fusion; i.e., contact between particles caused by the mixing and dilution is enough to allow the two dyes to become close enough to give rise to the FRET signal. This suggests that FRET sig-

nals produced during buffer exchange with ionizable lipids may also occur as a result of particle-particle contact, which occurs prior to complete particle fusion. Notably, the maximum FRET signal develops at pH values below the pKa of each ionizable lipid (e.g., pH 5.5 for MC3 formulations), although cryo-EM image (Figure 2B) indicates clearly that, for MC3 particles, considerable interaction between dense particles and liposome-like particles had already occurred at pH 5.5, well below the pKa of MC3, to form LNPs with bleb-like protrusions, allowing the DiO and DiI dyes to make contact. With regard to the FRET studies, it should be noted that they were carried out by diluting empty LNPs (i.e., mRNA was not included) 10-fold, to produce a final concentration of 1 mM total lipid. It is possible that particle contact, aggregation, fusion, and subsequent particle diameters are affected by concentration. We did not investigate effects of concentration in this study.

Our work suggests that the pKa of the ionizable lipid has a considerable effect during formation of LNPs and suggests that switching ionizable lipids without optimizing manufacturing protocols is likely to result in LNP products that differ in their morphology. LNPs produced with ionizable lipids with pKa of 6.10, 6.40, or 6.70, respectively, would remain 5%, 10%, or 20% protonated at pH 7.40. This difference would explain the SAXS data we obtained and perhaps suggests that it might be more appropriate to buffer finished SM-102 LNPs at pH 8.0, when they would be 5% protonated. The range of pKa values used in this study is limited (6.26 for MC3 to 6.70 for SM-102) because we chose to study the lipids used in successful products. It is clear from past reports that the optimum pKa of ionizable lipids is in the region of pKa 6.5, at least for delivery to hepatocytes after intravenous injection.¹⁷ It would be useful to extend the cryo-EM and SAXS studies to include lipids with a broader range of pKa values in future studies.

The fate of LNPs after uptake into the endocytic compartment of cells is important to consider and would be worthy of investigation. As the pH is progressively lowered during the processing of early and late endosomes,³⁴ it is evident that LNPs containing a lipid with a relatively high pKa, such as SM-102, would be proportionally more protonated earlier in the endocytic pathway, which would be expected to influence their fate and endosomal escape of their contents. This, coupled with their “less negative” zeta potential, which may influence cellular interaction, may explain why SM-102 LNPs produce higher levels of protein translation in our cell culture experiments.³⁵

Studies of the chemistry of ionizable lipids were reported in three substantial papers in 2010–2012,^{10,17,26} which ultimately led to the successful development of siRNA-LNP products. These two papers focused on understanding the roles of the ionizable head group and lipid tails. Head group pKa and tail shape were identified as critical determinants of biological function *in vivo*. The optimum pKa range of ionizable lipids, for knockdown of factor VII by siRNA delivered intravenously in mice, was 6.2–6.5.¹⁷ Lipids with pKa values above 6.5 were less effective in terms of factor VII knockdown, which contrasts with the higher biological activity *in vitro*,

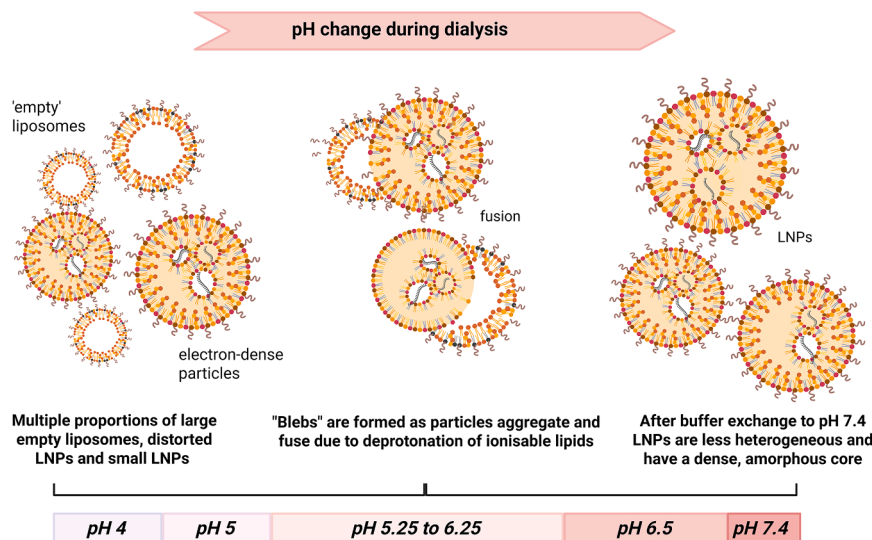


Figure 7. Schematic diagram of morphological changes during manufacture of mRNA-LNPs

mRNA-LNPs are generally formed initially at pH 4 in the presence of 25% ethanol. The removal of ethanol and buffer exchange by dialysis to produce the final product at physiological pH results in considerable structural rearrangement. Our observations suggest that LNPs produced using the standard formula of ionizable lipid/DSPC/cholesterol/DMG-PEG 2000 (mole ratio = 50/10/38.5/1.5) are initially polydisperse at pH 4 and contain mixed populations of particles including empty liposomes and particles containing mRNA. During buffer exchange, the LNPs form partially fused ("bleb") structures at intermediate pH values before condensing to form a more homogeneous population of particles typically with dense amorphous cores.

mentioned earlier. The optimum pKa is potentially specific to each application of LNPs and may be influenced by the effects of surface properties on biodistribution *in vivo*. Indeed, a study on intramuscular administration determined the optimum pKa range to be 6.6–6.9.³⁶ The study by Jayaraman et al.¹⁷ identified that MC3, later used in Onpattro, was one of the most effective of 50 lipids tested *in vivo*. MC3 has been referred to as a "second-generation" ionizable lipid, to distinguish it from "first-generation" lipids such as DODAP, which are less effective in LNP formulations.

Seiple et al. proposed that the shape of the lipid tail is also a critical determinant of biological function and suggested that a cone-shaped lipid with a low-volume head group at the tip could facilitate membrane fusion and endosome disruption.¹⁰ This philosophy led to the development of "third-generation" lipids, such as ALC-0315 and SM-102, which have branched lipid tails that are more cone shaped; i.e., the terminal regions of the lipid tails are wider than the head group regions. It is important to establish in what circumstances the branched lipids, used in the COVID-19 mRNA vaccines, result in higher activity and how the interplay between pKa and shape plays out *in vivo*. In the current study, the clearest difference in LNP morphology between the two branched lipids and MC3 or DODMA was evident in the lower d-spacing measured from the SAXS profiles. Linking the broad diffraction peaks in the SAXS profiles with bilayer-like structures at low pH, the branched lipids appeared to result in shorter distances between head groups within the bilayers. It will be interesting to investigate whether this explains the slightly higher percentage of endosome escape reported in our cell culture experiments.³⁵

In conclusion, our data show that pKa of the ionizable lipid is an important determinant of the morphology of fully formed LNPs after buffer exchange. During buffer exchange, our work suggests that intermediate, partially formed LNPs are stable for at least a week when held at intermediate pH values (i.e., pH 6.5). We suggest that lipid pKa may have a significant effect on the biodistribution of LNPs

and the fate of LNP after uptake into cells. In turn, pKa and LNP morphology will be a strong determinant of biological activity.

MATERIALS AND METHODS

Materials

The lipids 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC) and 1,2-dimyristoyl-rac-glycero-3-methoxypolyethylene glycol-2000 (DMG-PEG 2000) were purchased from Avanti Polar Lipids (Alabaster, AL). Cholesterol and DODMA were purchased from Sigma-Aldrich (St. Louis, MO). The ionizable lipids DLin-MC3-DMA (MC3), ALC-0315, and SM-102 were obtained from DC Chemicals (Shanghai, China). Sodium acetate, acetic acid, citric acid monohydrate, trisodium citrate dihydrate, sodium dihydrogen phosphate monohydrate, sodium dihydrogen phosphate dihydrate, Trizma hydrochloride buffer (1 M), and sucrose were obtained from Sigma-Aldrich (St. Louis, MO). Lipophilic indocarbocyanine dyes DiO and 1,1'-dioctadecyl3,3',3'-tetramethylindocarbocyanine perchlorate (DiI) were obtained from Thermo Fisher Scientific (Waltham, MA). The mRNA used in the study was from a large batch of an RBD-TM SARS-CoV-2 spike construct (Beta variant), originally prepared to manufacture a product for a phase 1 clinical trial. Details of this mRNA are published elsewhere.³⁷

Preparation of LNP-mRNA

LNPs were prepared using a microfluidics micromixer (NanoAssemblr benchtop or Ignite, from Precision Nanosystems, Vancouver, CA). A lipid stock solution of the ionizable lipid/DSPC/cholesterol/DMG-PEG 2000 in the ratio 50/10/38.5/1.5 mol% was prepared in ethanol and then mixed with a solution of mRNA in 25 mM acetate buffer, pH 4. The LNP particles were produced at a flow rate of 4 mL/min, using a 3:1 aqueous phase to ethanol phase flow rate ratio (v/v). Unless otherwise specified, the LNP particles were dialyzed overnight against 800 volumes of 25 mM acetate (pH 4) or 25 mM Tris buffer (pH 7.4) and formulated in 8.8% sucrose. When mRNA was included in the particles, we used an N/P ratio of 6.

Determination of pKa of ionizable lipids

pKa determination by TNS titration was performed using the LipidLaunch LNP apparent pKa assay kit (Cayman Chemical, MI, USA). Buffer solutions provided (pH 4.0 through 9.0) were diluted to 10 mM for pKa determination. 150 mM sodium chloride solution was added to adjust to physiological ionic strength. 8.3 μ L of 10 mM 6-(*p*-Toluidino)-2-naphthalenesulfonic acid sodium salt (TNS reagent) in DMSO was added to 992 μ L nucleic-free water to produce an 83 μ M TNS reagent solution. LNP was then prepared at a concentration equivalent to 500 μ M of each ionizable lipid in each formulations in 25 mM PBS buffer. 90 μ L of each buffer solution was mixed with 5 μ L LNP suspension and 5 μ L 83 μ M TNS reagent in triplicate in a black-bottomed 96-well plate. The plate was analyzed using a PerkinElmer EnVision plate reader using excitation and emission wavelengths of 320 and 450 nm, respectively. We used the LipidLaunch kit for convenience but measured the pH values of each buffer and used the experimental pH values to estimate pKa values. A sigmoidal plot of fluorescence versus buffer pH was determined. The apparent pKa of the lipid in each LNP formulation was determined as the pH at the mid-point of the inflection of the sigmoidal plot. pKa values for empty MC3 LNPs and mRNA-MC3-LNPs (N/P ratio = 6) were not significantly different.

FRET particle fusion experiments

The non-exchangeable FRET assay for indicating particle fusion and rearrangement of the LNPs was established in our laboratory based on its description by Kulkarni et al.¹¹ Empty LNP-DiO and LNP-DiI particles were formulated separately by including one of the lipophilic tracking dyes DiO (FRET donor) or DiI (FRET acceptor) in the lipid solution at 0.2 mol%. The LNPs were then dialyzed overnight against the pH 4 acetate buffer, resulting in dispersions containing 10 mM total lipid. LNP-DiO and LNP-DiI particles were mixed in a 1:1 ratio and then diluted 10-fold in a 100 mM buffer to give final pH values between 4 and 7.4. Samples were excited at 470 nm, and the emitted light was assayed over the range 505–640 nm. Spectroscopic evidence for FRET could be observed at 505 and 565 nm. The FRET intensity stabilized within 30 min of mixing and was analyzed within 4 h of mixing. Particle sizing was carried out within the same 4 h period.

Particle size and polydispersity determination

Particle size distribution and polydispersity were estimated by DLS and NTA. DLS measurements were performed using a Zetasizer Nano ZS (Malvern Panalytical, Malvern, UK). All measurements were performed three times for each technical replicate. NTA measurements were performed using a NanoSight NS300 (Malvern Panalytical, Malvern, UK) equipped with a scientific sCMOS camera, a 12x objective lens, a red laser module (642 nm), and NTA version 3.0 software. A 1 mL disposable syringe was used to inject the LNP samples into the instrument chamber. Every measurement was repeated five times. The FTLA concentration against the size graphs, the intensity against size graphs, and the 30 s NTA video data were collected for size distribution analysis.

SAXS studies

The SAXS experiments associated with this work were performed over multiple beamtime allocations on the SAXS/WAXS beamline at the Australian Synchrotron (ANSTO). Experiments were performed at the ambient temperature of the experimental hutch, which is typically 27°C. LNPs were stored at 4°C prior to analysis, within 3 days of preparation, and were physically stable during this period. Samples and associated buffers for background subtraction (100 μ L each) were loaded into 96-well PCR plates for sampling. An autosampler aspirated one sample or buffer background at a time into a 1.5 mm diameter capillary held in the X-ray beam (photon energy = 11.5 or 12.0 keV, wavelength λ = 1.078 or 1.033 Å) before returning the sample to the well from which it was drawn. Prior to each sample measurement, the appropriate buffer background was measured for background subtraction, and the capillary was automatically washed with Hellmanex detergent (2 vol% aqueous solution) and water before blowing dry with a stream of nitrogen after each sample measurement. Multiple 1-second acquisitions were collected as the samples were drawn into and expelled from the capillary. Images of the scattering pattern in 2D were recorded on a PILATUS 2M detector (Dectris, Switzerland) and reduced to plots of scattered X-ray intensity ($I(q)$) versus the magnitude of the scattering vector q ($= (4\pi/\lambda)\sin\theta$, where 2θ is the scattering angle) using software developed at the ANSTO (Scatterbrain). The sample-to-detector distance was set to 2,790 or 3,000 mm, and this afforded an observable q range between 0.005 and 0.5 Å⁻¹. Data were placed on an absolute scale (cm⁻¹) using scattering from water in the capillary as the standard before the scattering profile for each buffer background was subtracted from that for the corresponding sample. Data transformation and analysis were performed using Microsoft Excel or OriginLab Pro 2023 (OriginLab Corporation, Northampton, MA). Peak fitting of SAXS data and area under the curve was performed by peak analysis functionality in OriginLab Pro 2023. The corresponding repeating distances from the position of the broad diffraction peaks were calculated using $d = 2\pi/q$.

Cryo-EM

Cryo-EM allowed examination of LNP morphologies under different pH conditions. LNPs were stored at 4°C prior to analysis, within 3 days of preparation, and were physically stable during this period. Prior to imaging, an FEI Vitrobot Mark IV system at 4°C under 100% humidity was utilized for sample vitrification. Glow-discharged QuantiFoil lacey carbon grids were placed within the Vitrobot apparatus, and 3 μ L of LNP sample was applied. Samples were blotted for 3 s at a blot force of -3 and then plunged into liquid ethane before being transferred into liquid nitrogen for storage until use. LNPs were then imaged on a Tecnai Spirit 200 kV transmission electron microscope with Gatan Microscopy Suite operating software.

DATA AVAILABILITY

All data are available in the main text or the supplementary materials.

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used mRNA manufactured for production of a clinical batch of a COVID-19 vaccine to produce the mRNA-LNPs used in the study. We used infrastructure provided by Monash University and beamtime awarded to run SAXS experiments at the Australian Synchrotron. We also acknowledge Medical Research Future Fund (MRFF) mRNA Clinical Trials Enabling Infrastructure Award (MRFCT1000006) and Victorian Government (DJPR) – mRNA Victoria grant for “Development of a mRNA SARS-CoV-2 RBD-TM vaccine for Phase 1 clinical trial (2021–2022).”

AUTHOR CONTRIBUTIONS

Conceptualization: C.W.P., A.P.R.J., and Y.H.; methodology: Y.H., E.H.P., A.J.C., and N.M.K.; investigation: Y.H., E.H.P., W.-K.F., A.J.C., and N.M.K.; visualization: C.W.P., A.P.R.J., and Y.H.; funding acquisition: C.W.P., A.P.R.J., and A.J.C.; project administration: C.W.P.; supervision: C.W.P. and A.P.R.J.; writing: C.W.P. and Y.H.; writing – review of drafts: all authors.

DECLARATION OF INTERESTS

A provisional patent (PCT/AU2022/050912) assigned to Monash University specifies the design of the RBD-TM mRNA used in this study. C.W.P. is an inventor of this intellectual property.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.omtm.2025.101578>.

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